

Gamma Rays:

Gamma rays are **waves, not particles**. This means that they have **no mass** and **no charge**. Gamma rays have a **high penetrating power** - it takes a thick sheet of metal such as **lead**, or **concrete** to reduce them significantly. Gamma rays do not directly ionise other atoms, although they may cause atoms to emit other particles which will then cause ionisation.

Gamma rays (γ) are electromagnetic waves, rather like X rays and radio waves. Thus gamma rays have **no mass** and **no charge**.

After a nucleus has emitted an α -particle or a β -particle, it may still have too much energy: we say it is in an "excited state". It can get rid of this energy by emitting a pulse of very high frequency electromagnetic radiation, called a gamma ray. Gamma rays do not pull electrons off atoms they pass, as α -particles and β -particles do. This means that they do not lose much energy as they travel, as they do not interact as much with the matter they pass. Therefore, gamma rays have a **high penetrating power**, and a **very long range**.

It's worth noting that there is no such thing as a pure γ -ray source. Gamma rays are given off by most α -emitters and β -emitters. If we want a source of pure gamma rays, we can get it by using a substance that emits both β and γ , and simply keep it in an aluminium container that stops the β -particles.

Useful gamma sources include Technetium-99m, which is used as a "tracer" in medicine. This is a combined β and γ source, and is chosen because betas are less harmful to the patient than alphas (less ionisation) and because Technetium has a short half-life (just over 6 hours), so it decays away quickly and reduces the dose to the patient. In Gamma decay, both atomic number and atomic mass remain unchanged.

Detection of Radioactivity

Although some forms of electromagnetic energy, such as light and heat, can be detected by the human senses. One of the greatest drawbacks to high energy radiation is the inability to detect it. We cannot see, feel, taste, smell, or hear the various forms of ionizing radiation. Fortunately, ionizing radiation interacts with matter which makes detection and measurement possible by utilizing specialized equipment. In this section we want to introduce you to the various ways and means of detecting and measuring ionizing radiation.

Becquerel discovered radioactivity because it left marks on photographic film. However, there are more definitive means commonly used by scientists and technicians who study and work with radiation. The equipment utilized for the detection and measurement of radiation commonly employs some type of substance or material that responds to radiation. Many common methods use either an ionization process or molecular excitation process as a basis. Remember that radiation interacts with matter. For detection and measurement purposes the process of ionization is the most commonly employed technique, based on the principle of charged particles producing ion pairs by direct